

MATH1560J Assignment 7 - Sample Answer

Exercise 8.1

(a)

From the second equation, $t = y + 2$. Substitute into the first: $x = 1 - (y + 2)^2$

The direction follows as y increases.

(b)

$1 + x = y^2 \implies y^2 - x = 1 \implies x = y^2 - 1$, where $x \geq 0$.

The direction follows as x increases.

Exercise 8.2

(a)

$$\begin{aligned}\frac{dx}{dt} &= 2 \cos t, & \frac{dy}{dt} &= -3 \sin t \implies \frac{dy}{dx} = \frac{-3 \sin t}{2 \cos t} = -\frac{3}{2} \tan t \\ \frac{d^2y}{dx^2} &= \frac{\frac{d}{dt}(-\frac{3}{2} \tan t)}{\frac{dx}{dt}} = \frac{-\frac{3}{2} \sec^2 t}{2 \cos t} = -\frac{3}{4 \cos^3 t}\end{aligned}$$

Convex when $\frac{d^2y}{dx^2} > 0$: $-\frac{3}{4 \cos^3 t} > 0 \implies \cos t < 0 \implies t \in (\pi/2, 3\pi/2)$.

(b)

$$\begin{aligned}\frac{dx}{dt} &= -2 \sin 2t, & \frac{dy}{dt} &= -\sin t \implies \frac{dy}{dx} = \frac{-\sin t}{-4 \sin t \cos t} = \frac{1}{4 \cos t} = \frac{1}{4} \sec t \\ \frac{d^2y}{dx^2} &= \frac{\frac{d}{dt}(\frac{1}{4} \sec t)}{-2 \sin 2t} = \frac{\frac{1}{4} \sec t \tan t}{-4 \sin t \cos t} = \frac{\frac{\sin t}{4 \cos^2 t}}{-4 \sin t \cos t} = -\frac{1}{16 \cos^3 t}\end{aligned}$$

Convex when $\frac{d^2y}{dx^2} > 0$: $-\frac{1}{16 \cos^3 t} > 0 \implies \cos t < 0 \implies t \in (\pi/2, \pi)$.

Exercise 8.3

(a)

$$\text{Area} A = \int_{\alpha}^{\beta} y(\theta) x'(\theta) d\theta = \int_{2\pi}^0 a \sin^3 \theta (3a \cos^2 \theta (-\sin \theta)) d\theta = 3a^2 \int_0^{2\pi} \sin^4 \theta - \sin^6 \theta d\theta = \frac{3a^2}{8} \pi$$

(b)

$$\begin{aligned}\text{Length} &= \int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\&= \int_0^{2\pi} \sqrt{(-3a \cos^2 \theta \sin \theta)^2 + (3a \sin^2 \theta \cos \theta)^2} d\theta \\&= \int_0^{2\pi} 3a |\sin \theta \cos \theta| d\theta \\&= \frac{3}{2}a \int_0^{2\pi} |\sin(2\theta)| d\theta \\&= 6a \left[-\frac{1}{2} \cos(2\theta)\right]_0^{\pi/2} \\&= 6a\end{aligned}$$

Exercise 8.4

(a)

$$\frac{dx}{d\theta} = -\sin \theta \cos \theta - (1 + \cos \theta) \sin \theta = -\sin \theta (1 + 2 \cos \theta),$$

$$\frac{dy}{d\theta} = -\sin^2 \theta + (1 + \cos \theta) \cos \theta = \cos \theta + \cos 2\theta.$$

Horizontal tangents:

$$\frac{dy}{d\theta} = 0 \implies \cos \theta + \cos 2\theta = 0 \implies 2 \cos^2 \theta + \cos \theta - 1 = 0 \implies \cos \theta = \frac{1}{2}, -1.$$

Thus $\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \pi$

At $\theta = \pi$, $\frac{dx}{d\theta} = 0$, so this point is excluded.

Hence horizontal tangents occur at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$

Vertical tangents:

$$\frac{dx}{d\theta} = 0 \implies \sin \theta (1 + 2 \cos \theta) = 0.$$

Thus $\theta = 0, \pi, \frac{2\pi}{3}, \frac{4\pi}{3}$

At $\theta = \pi$, $\frac{dy}{d\theta} = 0$, so it is excluded.

Vertical tangents occur at $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$

(b)

$$\frac{dy}{d\theta} = e^\theta \sin \theta + e^\theta \cos \theta = e^\theta (\sin \theta + \cos \theta)$$

$$\frac{dx}{d\theta} = e^\theta \cos \theta - e^\theta \sin \theta = e^\theta (\cos \theta - \sin \theta)$$

Horizontal tangents:

$$\frac{dy}{d\theta} = 0 \implies e^\theta (\sin \theta + \cos \theta) = 0 \implies \tan \theta = -1 \implies \theta = -\frac{\pi}{4} + k\pi, k \in \mathbb{Z}$$

Vertical tangents:

$$\frac{dx}{d\theta} = 0 \implies e^\theta (\cos \theta - \sin \theta) = 0 \implies \tan \theta = 1 \implies \theta = \frac{\pi}{4} + k\pi, k \in \mathbb{Z}$$

Exercise 8.5

Loop is formed when $r = 0 \implies 2 \cos^2 \theta = 1 \implies \cos \theta = \pm \frac{1}{\sqrt{2}}$ Interval: $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$

$$A = \frac{1}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (2 \cos \theta - \sec \theta)^2 d\theta = \int_0^{\frac{\pi}{4}} (2 \cos 2\theta - 2 + \sec^2 \theta) d\theta = [\sin 2\theta - 2\theta + \tan \theta]_0^{\frac{\pi}{4}} = 2 - \frac{\pi}{2}$$

Exercise 8.6

(a)

Intersection: $2 \cos \theta = 1 \implies \theta = \pm \frac{\pi}{3}$

$$A = \frac{1}{2} \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} ((2 \cos \theta)^2 - 1^2) d\theta = \int_0^{\frac{\pi}{3}} (4 \cos^2 \theta - 1) d\theta = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

(Or, notice that the curve $r = \cos 2\theta$ is the circle $(x-1)^2 + y^2 = 1$, $r = 1$ is the circle $x^2 + y^2 = 1$)

(b)

Intersection: $1 - \sin \theta = 1 \implies \sin \theta = 0 \implies \theta = 0, \pi$.

Inside $r = 1 - \sin \theta$ outside $r = 1$ occurs when $1 - \sin \theta > 1 \implies \sin \theta < 0 \implies \theta \in (\pi, 2\pi)$.

$$A = \frac{1}{2} \int_{\pi}^{2\pi} ((1 - \sin \theta)^2 - 1^2) d\theta = \frac{1}{2} \int_{\pi}^{2\pi} (-2 \sin \theta + \frac{1 - \cos 2\theta}{2}) d\theta = \frac{1}{2} [2 \cos \theta + \frac{\theta}{2} - \frac{\sin 2\theta}{4}]_{\pi}^{2\pi} = 2 + \frac{\pi}{4}$$

Exercise 8.7

(a)

$$L = \int_0^{\pi} \sqrt{(2 \cos \theta)^2 + (-2 \sin \theta)^2} d\theta = \int_0^{\pi} \sqrt{4(\cos^2 \theta + \sin^2 \theta)} d\theta = \int_0^{\pi} 2 d\theta = [2\theta]_0^{\pi} = 2\pi$$

(b)

$$L = \int_0^{2\pi} \sqrt{(5^\theta)^2 + (5^\theta \ln 5)^2} d\theta = \int_0^{2\pi} 5^\theta \sqrt{1 + (\ln 5)^2} d\theta = \sqrt{1 + (\ln 5)^2} \left[\frac{5^\theta}{\ln 5} \right]_0^{2\pi} = \frac{\sqrt{1 + (\ln 5)^2}}{\ln 5} (5^{2\pi} - 1)$$

(c)

$$L = \int_0^{2\pi} \sqrt{(\theta^2)^2 + (2\theta)^2} d\theta = \int_0^{2\pi} \sqrt{\theta^4 + 4\theta^2} d\theta = \int_0^{2\pi} \theta \sqrt{\theta^2 + 4} d\theta$$

Let $u = \theta^2 + 4$, $du = 2\theta d\theta$

$$L = \frac{1}{2} \int_4^{4\pi^2+4} \sqrt{u} du = \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_4^{4\pi^2+4} = \frac{1}{3} \left((4\pi^2 + 4)^{\frac{3}{2}} - 4^{\frac{3}{2}} \right) = \frac{8}{3} \left((\pi^2 + 1)^{\frac{3}{2}} - 1 \right)$$

(d)

$$L = \int_0^{2\pi} \sqrt{4(1 + \cos \theta)^2 + 4 \sin^2 \theta} d\theta = \int_0^{2\pi} \sqrt{8(1 + \cos \theta)} d\theta = \int_0^{2\pi} \sqrt{16 \cos^2(\frac{\theta}{2})} d\theta = \int_0^{2\pi} 4 |\cos(\frac{\theta}{2})| d\theta = 16$$

Exercise 8.8

(a)

If (a, b) lies on the curve for $t = t_0$, then $a = \frac{3t_0}{1+t_0^3}$ and $b = \frac{3t_0^2}{1+t_0^3}$. Let $t = 1/t_0$:

$$x(1/t_0) = \frac{3/t_0}{1+1/t_0^3} = \frac{3t_0^2}{t_0^3+1} = b, \quad y(1/t_0) = \frac{3/t_0^2}{1+1/t_0^3} = \frac{3t_0}{t_0^3+1} = a$$

Thus, the curve is symmetric with respect to $y = x$.

Intersection with $y = x$:

$$\frac{3t^2}{1+t^3} = \frac{3t}{1+t^3} \implies 3t^2 = 3t \implies t = 1, 0$$

Point: $(x(1), y(1)) = (\frac{3}{2}, \frac{3}{2})$, $(x(0), y(0)) = (0, 0)$

(b)

$$\frac{dx}{dt} = \frac{3(1-2t^3)}{(1+t^3)^2}, \quad \frac{dy}{dt} = \frac{3t(2-t^3)}{(1+t^3)^2} \implies \frac{dy}{dx} = \frac{t(2-t^3)}{1-2t^3}$$

Horizontal tangents:

$$\frac{dy}{dt} = 0, \frac{dx}{dt} \neq 0 \implies t = 0, t = \sqrt[3]{2} \implies (0, 0) \text{ and } (\sqrt[3]{2}, \sqrt[3]{4})$$

Vertical tangents:

$$\frac{dx}{dt} = 0, \frac{dy}{dt} \neq 0: t = \sqrt[3]{1/2} \implies (\sqrt[3]{4}, \sqrt[3]{2}).$$

(c)

As $t \rightarrow -1$:

$$x + y + 1 = \frac{3t + 3t^2 + 1 + t^3}{1 + t^3} = \frac{(t+1)^3}{(t+1)(t^2 - t + 1)} = \frac{(t+1)^2}{t^2 - t + 1}$$

$$\lim_{t \rightarrow -1} (x + y + 1) = \frac{0}{3} = 0 \implies y = -x - 1.$$

(e)

$$x^3 + y^3 = \frac{27t^3 + 27t^6}{(1+t^3)^3} = \frac{27t^3(1+t^3)}{(1+t^3)^3} = \frac{27t^3}{(1+t^3)^2}$$
$$3xy = 3 \left(\frac{3t}{1+t^3} \right) \left(\frac{3t^2}{1+t^3} \right) = \frac{27t^3}{(1+t^3)^2}$$

Thus, $x^3 + y^3 = 3xy$.

(f)

Substitute $x = r \cos \theta, y = r \sin \theta$ into $x^3 + y^3 = 3xy$:

$$\begin{aligned} r^3 \cos^3 \theta + r^3 \sin^3 \theta &= 3r^2 \cos \theta \sin \theta \\ r &= \frac{3 \cos \theta \sin \theta}{\cos^3 \theta + \sin^3 \theta} = \frac{3 \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\cos \theta}}{1 + \frac{\sin^3 \theta}{\cos^3 \theta}} = \frac{3 \tan \theta \sec \theta}{1 + \tan^3 \theta} \end{aligned}$$

(g)

Using $A = \frac{1}{2} \int (x\dot{y} - y\dot{x})dt$:

$$\begin{aligned} x\dot{y} - y\dot{x} &= \frac{9t^2}{(1+t^3)^2} \\ A &= \frac{1}{2} \int_0^\infty \frac{9t^2}{(1+t^3)^2} dt = \frac{3}{2} \left[-\frac{1}{1+t^3} \right]_0^\infty = \frac{3}{2} \end{aligned}$$

(h)

$$\text{Curve : } r_c(\theta) = \frac{3 \cos \theta \sin \theta}{\cos^3 \theta + \sin^3 \theta}, \quad \text{Asymptote : } r_L(\theta) = \frac{-1}{\cos \theta + \sin \theta}$$

The area A_{ext} between the branches and the asymptote for $\theta \in (-\frac{\pi}{4}, 0)$ (in the second and fourth quadrant) and its symmetric part:

$$A_{ext} = 2 \cdot \frac{1}{2} \int_{-\frac{\pi}{4}}^0 (r_L^2(\theta) - r_c^2(\theta)) d\theta = \int_{-\frac{\pi}{4}}^0 \left[\frac{1}{(\cos \theta + \sin \theta)^2} - \frac{9 \sin^2 \theta \cos^2 \theta}{(\cos^3 \theta + \sin^3 \theta)^2} \right] d\theta$$

Let $u = \tan \theta, du = \sec^2 \theta d\theta$:

$$\begin{aligned} \frac{1}{(\cos \theta + \sin \theta)^2} d\theta &= \frac{\sec^2 \theta}{(1 + \tan \theta)^2} d\theta = \frac{du}{(1+u)^2} \\ \frac{9 \sin^2 \theta \cos^2 \theta}{(\cos^3 \theta + \sin^3 \theta)^2} d\theta &= \frac{9 \tan^2 \theta \sec^2 \theta}{(1 + \tan^3 \theta)^2} d\theta = \frac{9u^2 du}{(1+u^3)^2} \end{aligned}$$

The integral becomes:

$$\begin{aligned} A_{ext} &= \int_{-1}^0 \left[\frac{1}{(1+u)^2} - \frac{9u^2}{(1+u^3)^2} \right] du = \left[-\frac{1}{1+u} - \left(-\frac{3}{1+u^3} \right) \right]_{-1}^0 \\ A_{ext} &= F(0) - \lim_{u \rightarrow -1^+} F(u) = \frac{2-0}{1-0+0} - \frac{2-(-1)}{1-(-1)+(-1)^2} = 2 - \frac{3}{3} = 1 \end{aligned}$$

Add the triangle in the third quadrant:

$$A_{ext} = 1 + \frac{1}{2} \cdot 1 \cdot 1 = \frac{3}{2}$$

Since $A_{loop} = \frac{3}{2}$ from (g), we have $A_{loop} = A_{ext}$.